S-RQ3. Supplementary Material for RQ3 (Fault Detection)

This appendix collects the supplementary tables referenced in the RQ3 results section (Section 7.3) of the main manuscript. Tables S1–S6 give the full descriptive statistics for both mutation operators at all coverage levels reported in the article. Tables S7 and S8 give the full pairwise Wilcoxon results that back the cross-approach claims. Table S9 breaks the Edge–Event paired comparisons out by individual (SPL, approach) combination, with median values, paired delta, p_{BH}, and A_{12} effect size; the manuscript table reports SPL-level aggregate counts only. Tables S10 and S11 report the damping-factor sensitivity and the multi-seed stability analyses in full. Table S12 gives the fault-detection results obtained from the original RQ1 single-seed Random Walk sequences (the RQ3^a test-suite set in Table **??**); the corresponding manuscript results use the multi-seed median (RQ3^b).

All percentage values are rounded to one decimal place using ROUND_HALF_UP after a six-decimal float-noise scrub. The syngo.via structural baseline value at $L = 2$ is shown to two decimal places for consistency with the precision used elsewhere in the supplementary data files.

Table 1
S1: Median mutation score on edge omission mutants (%), per approach, L level, and SPL. Stochastic baseline rows aggregate the per-product median across the 10 seeds (RQ3^b) and then the median across products.

Approach	L	SVM	eM	EI	BAv2	SAS	Svia	Te	HS
Model Once, Generate Any	2	100	100	100	100	100	100	100	100
	3	100	100	100	100	100	100	100	100
	4	100	95.2	100	91.5	100	100	100	100
Structural baseline	2	100	35.3	100	100	66.7	3.57	100	40.3
	3	100	76.2	100	100	85.3	100	100	41.7
	4	100	82.4	100	100	86.4	100	100	43.1
Stochastic baseline	—	100	100	100	100	100	100	100	100

Table 2
S2: Median mutation score on event omission mutants (%). Same layout and aggregation as Table 1.

Approach	L	SVM	eM	EI	BAv2	SAS	Svia	Te	HS
Model Once, Generate Any	2	100	100	100	100	100	100	100	100
	3	100	100	100	100	100	100	100	100
	4	100	85.7	100	87	100	100	100	100
Structural baseline	2	100	80	100	100	92.9	8.56	100	33.3
	3	100	100	100	100	97.5	100	100	35
	4	100	100	100	100	100	100	100	36.7
Stochastic baseline	—	100	100	100	100	100	100	100	100

Table 3
S3: Median penalized percent of suite to detect on edge omission mutants (%). Mutants the suite fails to kill are charged a 100% cost (Section **??**); rows of 100.0 reflect pervasive mutant survival rather than late detection.

Approach	L	SVM	eM	El	BAv2	SAS	Svia	Te	HS
	2	58.1	54.6	49	47.1	39.3	2	11.4	5.6
Model Once, Generate Any	3	58.1	21.1	19.6	44.5	22.8	2.4	8.5	3.7
	4	58.1	12.1	12.3	50	14	2	8.5	3.2
	2	31.7	100	50	35.6	60.1	100.00	6.8	100
Structural baseline	3	38.4	60.3	14.1	38.5	31.4	7.6	5.7	100
	4	43	24.6	10	41.4	22.2	2.5	4.9	100
Stochastic baseline	—	64.3	25.2	19.6	46.8	22.1	0.2	6.5	2.8

Table 4
S4: Median penalized percent of suite to detect on event omission mutants (%). Same layout as Table 3.

Approach	L	SVM	eM	El	BAv2	SAS	Svia	Te	HS
	2	47.7	26.8	17.1	37.9	23.7	2	9.2	5.6
Model Once, Generate Any	3	47.7	31.9	7.4	41.9	13.2	2.4	6.8	3.7
	4	47.7	13.6	5.7	51.2	9	2	6.8	3.2
	2	27.8	52.9	12.2	31.5	27.5	100.00	3.9	100
Structural baseline	3	31.3	26.9	4.6	34.3	18.5	7.6	3.6	100
	4	35.3	13.6	3.6	38	13.9	2.5	2.7	100
Stochastic baseline	—	57.1	13.6	5.6	43.9	15.2	0.2	4.9	2.8

Table 5
S5: Median *unpenalized* percent of suite to detect on edge omission mutants (%). Computed only over mutants the suite kills and excludes failures from the average. Compare with Table 3: the two coincide wherever the mutation score reaches 100%, but the unpenalized variant is much lower for the structural baseline on eMail, syngo.via, and Hockerty Shirts because it discards the unkilled mutants from the average.

Approach	L	SVM	eM	El	BAv2	SAS	Svia	Te	HS
	2	58.1	54.6	49	47.1	39.3	2	11.4	5.6
Model Once, Generate Any	3	58.1	21.1	19.6	44.5	22.8	2.4	8.5	3.7
	4	58.1	10.8	12.3	46.7	14	2	8.5	3.2
	2	31.7	47.1	49	35.3	28.3	33.98	6.8	7.6
Structural baseline	3	38.4	44.8	14.1	37.8	24.4	7.6	5.7	4.7
	4	43	19.4	10	40.9	18.2	2.5	4.9	3.9
Stochastic baseline	—	64.3	25.2	19.6	46.8	22.1	0.2	6.5	2.8

Table 6
S6: Median *unpenalized* percent of suite to detect on event omission mutants (%). Same caveat as Table 5.

Approach	L	SVM	eM	El	BAv2	SAS	Svia	Te	HS
	2	47.7	26.8	17.1	37.9	23.7	2	9.2	5.6
Model Once, Generate Any	3	47.7	31.9	7.4	41.9	13.2	2.4	6.8	3.7
	4	47.7	10.5	5.7	47	9	2	6.8	3.2
	2	27.8	38.5	12.2	31.3	25.4	18.45	3.9	3.8
Structural baseline	3	31.3	25.9	4.6	34	18.1	7.6	3.6	2.8
	4	35.3	13.6	3.6	38	13.9	2.5	2.7	2.3
Stochastic baseline	—	57.1	13.6	5.6	43.9	15.2	0.2	4.9	2.8

Table 7
S7: Pairwise Wilcoxon signed-rank results on edge omission (Benjamini–Hochberg corrected, $\alpha = 0.05$). MO denotes *Model Once, Generate Any* and Struct denotes the structural baseline. The winner column reports which side of each comparison is favoured by the Vargha–Delaney A_{12} value.

Metric	SPL	Approach A	Approach B	n	W	p_{BH}	A_{12}	Winner
MS	SVM	Struct L=2	Stoch	12	0	—	0.50	tie
	SVM	Struct L=3	Stoch	12	0	—	0.50	tie
	SVM	Struct L=4	Stoch	12	0	—	0.50	tie
	SVM	MO L=2	Struct L=2	12	0	—	0.50	tie
	SVM	MO L=2	Stoch	12	0	—	0.50	tie
	SVM	MO L=3	Struct L=3	12	0	—	0.50	tie
	SVM	MO L=3	Stoch	12	0	—	0.50	tie
	SVM	MO L=4	Struct L=4	12	0	—	0.50	tie
	SVM	MO L=4	Stoch	12	0	—	0.50	tie
	eM	Struct L=2	Stoch	23	0	—	0.00	B
	eM	Struct L=3	Stoch	23	0	—	0.00	B
	eM	Struct L=4	Stoch	23	0	—	0.00	B
	eM	MO L=2	Struct L=2	23	0	—	1.00	A
	eM	MO L=2	Stoch	23	0	—	0.50	tie
	eM	MO L=3	Struct L=3	23	0	—	1.00	A
	eM	MO L=3	Stoch	23	0	—	0.50	tie
	eM	MO L=4	Struct L=4	23	0	—	1.00	A
	eM	MO L=4	Stoch	23	0	—	0.17	B
	EI	Struct L=2	Stoch	42	0	—	0.36	B
	EI	Struct L=3	Stoch	42	0	—	0.50	tie
	EI	Struct L=4	Stoch	42	0	—	0.50	tie
	EI	MO L=2	Struct L=2	42	0	—	0.64	A
	EI	MO L=2	Stoch	42	0	—	0.50	tie
	EI	MO L=3	Struct L=3	42	0	—	0.50	tie
	EI	MO L=3	Stoch	42	0	—	0.50	tie
	EI	MO L=4	Struct L=4	42	0	—	0.50	tie
	EI	MO L=4	Stoch	42	0	—	0.50	tie
	BAv2	Struct L=2	Stoch	498	0	—	0.43	B
	BAv2	Struct L=3	Stoch	498	0	—	0.43	B
	BAv2	Struct L=4	Stoch	498	0	—	0.43	B
	BAv2	MO L=2	Struct L=2	498	0	—	0.57	A
	BAv2	MO L=2	Stoch	498	0	—	0.50	tie
	BAv2	MO L=3	Struct L=3	498	0	—	0.57	A
	BAv2	MO L=3	Stoch	498	0	—	0.50	tie
	BAv2	MO L=4	Struct L=4	498	0	—	0.11	B
	BAv2	MO L=4	Stoch	498	0	—	0.05	B
	SAS	Struct L=2	Stoch	2664	0	—	0.00	B
	SAS	Struct L=3	Stoch	2664	0	—	0.00	B
	SAS	Struct L=4	Stoch	2664	0	—	0.00	B
	SAS	MO L=2	Struct L=2	2664	0	—	1.00	A
	SAS	MO L=2	Stoch	2664	0	—	0.50	tie
	SAS	MO L=3	Struct L=3	2664	0	—	1.00	A
	SAS	MO L=3	Stoch	2664	0	—	0.50	tie
	SAS	MO L=4	Struct L=4	2664	0	—	1.00	A
	SAS	MO L=4	Stoch	2664	0	—	0.50	tie
	Svia	Struct L=2	Stoch	400	0	—	0.00	B
	Svia	Struct L=3	Stoch	400	0	—	0.50	tie
	Svia	Struct L=4	Stoch	400	0	—	0.50	tie
	Svia	MO L=2	Struct L=2	400	0	—	1.00	A
	Svia	MO L=2	Stoch	400	0	—	0.50	tie
	Svia	MO L=3	Struct L=3	400	0	—	0.50	tie
	Svia	MO L=3	Stoch	400	0	—	0.50	tie
	Svia	MO L=4	Struct L=4	400	0	—	0.50	tie
	Svia	MO L=4	Stoch	400	0	—	0.50	tie
	Te	Struct L=2	Stoch	400	0	—	0.50	tie
	Te	Struct L=3	Stoch	400	0	—	0.50	tie
	Te	Struct L=4	Stoch	400	0	—	0.50	tie
	Te	MO L=2	Struct L=2	400	0	—	0.50	tie
	Te	MO L=2	Stoch	400	0	—	0.50	tie
	Te	MO L=3	Struct L=3	400	0	—	0.50	tie
	Te	MO L=3	Stoch	400	0	—	0.50	tie
	Te	MO L=4	Struct L=4	400	0	—	0.50	tie
	Te	MO L=4	Stoch	400	0	—	0.50	tie
	HS	Struct L=2	Stoch	416	0	—	0.09	B
	HS	Struct L=3	Stoch	416	0	—	0.09	B
	HS	Struct L=4	Stoch	416	0	—	0.09	B

Table 8
S8: Pairwise Wilcoxon signed-rank results on event omission (Benjamini–Hochberg corrected, $\alpha = 0.05$). Same conventions as Table 7.

Metric	SPL	Approach A	Approach B	n	W	p_{BH}	A_{12}	Winner
MS	SVM	Struct L=2	Stoch	12	0	—	0.50	tie
	SVM	Struct L=3	Stoch	12	0	—	0.50	tie
	SVM	Struct L=4	Stoch	12	0	—	0.50	tie
	SVM	MO L=2	Struct L=2	12	0	—	0.50	tie
	SVM	MO L=2	Stoch	12	0	—	0.50	tie
	SVM	MO L=3	Struct L=3	12	0	—	0.50	tie
	SVM	MO L=3	Stoch	12	0	—	0.50	tie
	SVM	MO L=4	Struct L=4	12	0	—	0.50	tie
	SVM	MO L=4	Stoch	12	0	—	0.50	tie
	eM	Struct L=2	Stoch	23	0	—	0.04	B
	eM	Struct L=3	Stoch	23	0	—	0.33	B
	eM	Struct L=4	Stoch	23	0	—	0.50	tie
	eM	MO L=2	Struct L=2	23	0	—	0.96	A
	eM	MO L=2	Stoch	23	0	—	0.50	tie
	eM	MO L=3	Struct L=3	23	0	—	0.67	A
	eM	MO L=3	Stoch	23	0	—	0.50	tie
	eM	MO L=4	Struct L=4	23	0	—	0.17	B
	eM	MO L=4	Stoch	23	0	—	0.17	B
	EI	Struct L=2	Stoch	42	0	—	0.50	tie
	EI	Struct L=3	Stoch	42	0	—	0.50	tie
	EI	Struct L=4	Stoch	42	0	—	0.50	tie
	EI	MO L=2	Struct L=2	42	0	—	0.50	tie
	EI	MO L=2	Stoch	42	0	—	0.50	tie
	EI	MO L=3	Struct L=3	42	0	—	0.50	tie
	EI	MO L=3	Stoch	42	0	—	0.50	tie
	EI	MO L=4	Struct L=4	42	0	—	0.50	tie
	EI	MO L=4	Stoch	42	0	—	0.50	tie
	BAv2	Struct L=2	Stoch	498	0	—	0.43	B
	BAv2	Struct L=3	Stoch	498	0	—	0.43	B
	BAv2	Struct L=4	Stoch	498	0	—	0.50	tie
	BAv2	MO L=2	Struct L=2	498	0	—	0.57	A
	BAv2	MO L=2	Stoch	498	0	—	0.50	tie
	BAv2	MO L=3	Struct L=3	498	0	—	0.57	A
	BAv2	MO L=3	Stoch	498	0	—	0.50	tie
	BAv2	MO L=4	Struct L=4	498	0	—	0.05	B
	BAv2	MO L=4	Stoch	498	0	—	0.05	B
	SAS	Struct L=2	Stoch	2664	0	—	0.03	B
	SAS	Struct L=3	Stoch	2664	0	—	0.24	B
	SAS	Struct L=4	Stoch	2664	0	—	0.50	tie
	SAS	MO L=2	Struct L=2	2664	0	—	0.97	A
	SAS	MO L=2	Stoch	2664	0	—	0.50	tie
	SAS	MO L=3	Struct L=3	2664	0	—	0.76	A
	SAS	MO L=3	Stoch	2664	0	—	0.50	tie
	SAS	MO L=4	Struct L=4	2664	0	—	0.50	tie
	SAS	MO L=4	Stoch	2664	0	—	0.50	tie
	Svia	Struct L=2	Stoch	400	0	—	0.00	B
	Svia	Struct L=3	Stoch	400	0	—	0.50	tie
	Svia	Struct L=4	Stoch	400	0	—	0.50	tie
	Svia	MO L=2	Struct L=2	400	0	—	1.00	A
	Svia	MO L=2	Stoch	400	0	—	0.50	tie
	Svia	MO L=3	Struct L=3	400	0	—	0.50	tie
	Svia	MO L=3	Stoch	400	0	—	0.50	tie
	Svia	MO L=4	Struct L=4	400	0	—	0.50	tie
	Svia	MO L=4	Stoch	400	0	—	0.50	tie
	Te	Struct L=2	Stoch	400	0	—	0.50	tie
	Te	Struct L=3	Stoch	400	0	—	0.50	tie
	Te	Struct L=4	Stoch	400	0	—	0.50	tie
	Te	MO L=2	Struct L=2	400	0	—	0.50	tie
	Te	MO L=2	Stoch	400	0	—	0.50	tie
	Te	MO L=3	Struct L=3	400	0	—	0.50	tie
	Te	MO L=3	Stoch	400	0	—	0.50	tie
	Te	MO L=4	Struct L=4	400	0	—	0.50	tie
	Te	MO L=4	Stoch	400	0	—	0.50	tie
	HS	Struct L=2	Stoch	416	0	—	0.09	B
	HS	Struct L=3	Stoch	416	0	—	0.09	B
	HS	Struct L=4	Stoch	416	0	—	0.09	B
	HS	MO L=2	Struct L=2	416	0	—	0.91	A

Table 9
S9: Per-(SPL, approach) detail of the Edge–Event paired Wilcoxon tests reported in aggregate in Table **??**. Each row corresponds to one of the 56 SPL×approach combinations: 8 SPLs times the 7 approaches reported in this article. Median edge / event values, paired delta (event minus edge), Benjamini–Hochberg corrected p-value, and Vargha–Delaney A_{12} are listed for both the subsumption (mutation score) and earliness (penalized DC) tests. Approach abbreviations: MO = *Model Once, Generate Any*, Struct = structural baseline, Stoch = stochastic baseline.

			Subsumption (mutation score, %)					**Earliness (penalized DC, %)**				
SPL	**Approach**	n	**Edge**	**Event**	Δ	p_{BH}	A_{12}	**Edge**	**Event**	Δ	p_{BH}	A_{12}
SVM	MO L=2	12	100	100	0.0	n.s.	0.50	58.1	47.7	-7.2	<0.001	0.27
	MO L=3	12	100	100	0.0	n.s.	0.50	58.1	47.7	-7.2	<0.001	0.27
	MO L=4	12	100	100	0.0	n.s.	0.50	58.1	47.7	-7.2	<0.001	0.27
	Struct L=2	12	100	100	0.0	n.s.	0.50	31.7	27.8	-3.6	<0.001	0.22
	Struct L=3	12	100	100	0.0	n.s.	0.50	38.4	31.3	-7.1	<0.001	0.33
	Struct L=4	12	100	100	0.0	n.s.	0.50	43	35.3	-7.0	<0.001	0.36
	Stoch	12	100	100	0.0	n.s.	0.50	64.3	57.1	-7.2	<0.001	0.22
eM	MO L=2	23	100	100	0.0	n.s.	0.50	54.6	26.8	-28.6	<0.001	0.00
	MO L=3	23	100	100	0.0	n.s.	0.50	21.1	31.9	8.0	<0.001	0.78
	MO L=4	23	95.2	85.7	-9.7	0.002	0.29	12.1	13.6	0.9	n.s.	0.61
	Struct L=2	23	35.3	80	43.4	<0.001	1.00	100	52.9	-47.1	<0.001	0.00
	Struct L=3	23	76.2	100	21.7	<0.001	1.00	60.3	26.9	-31.3	<0.001	0.05
	Struct L=4	23	82.4	100	17.7	<0.001	1.00	24.6	13.6	-12.2	<0.001	0.14
	Stoch	23	100	100	0.0	n.s.	0.50	24.1	15.4	-8.5	<0.001	0.20
EI	MO L=2	42	100	100	0.0	n.s.	0.50	49	17.1	-29.1	<0.001	0.00
	MO L=3	42	100	100	0.0	n.s.	0.50	19.6	7.4	-12.2	<0.001	0.00
	MO L=4	42	100	100	0.0	n.s.	0.50	12.3	5.7	-6.4	<0.001	0.00
	Struct L=2	42	100	100	0.0	0.006	0.64	50	12.2	-36.5	<0.001	0.00
	Struct L=3	42	100	100	0.0	n.s.	0.50	14.1	4.6	-10.0	<0.001	0.00
	Struct L=4	42	100	100	0.0	n.s.	0.50	10	3.6	-7.7	<0.001	0.01
	Stoch	42	100	100	0.0	n.s.	0.50	18.4	5.1	-12.9	<0.001	0.01
BAv2	MO L=2	498	100	100	0.0	n.s.	0.50	47.1	37.9	-8.9	<0.001	0.07
	MO L=3	498	100	100	0.0	n.s.	0.50	44.5	41.9	-2.8	<0.001	0.39
	MO L=4	498	91.5	87	-4.1	<0.001	0.30	50	51.2	0.0	n.s.	0.54
	Struct L=2	498	100	100	0.0	<0.001	0.51	35.6	31.5	-3.6	<0.001	0.31
	Struct L=3	498	100	100	0.0	<0.001	0.51	38.5	34.3	-3.3	<0.001	0.36
	Struct L=4	498	100	100	0.0	<0.001	0.57	41.4	38	-3.2	<0.001	0.37
	Stoch	498	100	100	0.0	n.s.	0.50	47	44.1	-3.1	<0.001	0.34
SAS	MO L=2	2664	100	100	0.0	n.s.	0.50	39.3	23.7	-15.1	<0.001	0.00
	MO L=3	2664	100	100	0.0	n.s.	0.50	22.8	13.2	-9.4	<0.001	0.01
	MO L=4	2664	100	100	0.0	n.s.	0.50	14	9	-5.1	<0.001	0.09
	Struct L=2	2664	66.7	92.9	25.6	<0.001	0.98	60.1	27.5	-31.5	<0.001	0.02
	Struct L=3	2664	85.3	97.5	13.2	<0.001	1.00	31.4	18.5	-11.9	<0.001	0.02
	Struct L=4	2664	86.4	100	13.6	<0.001	1.00	22.2	13.9	-7.9	<0.001	0.05
	Stoch	2664	100	100	0.0	n.s.	0.50	21.7	15.1	-6.1	<0.001	0.18
Svia	MO L=2	400	100	100	0.0	n.s.	0.50	2	2	0.0	n.s.	0.50
	MO L=3	400	100	100	0.0	n.s.	0.50	2.4	2.4	0.0	n.s.	0.50
	MO L=4	400	100	100	0.0	n.s.	0.50	2	2	0.0	n.s.	0.50
	Struct L=2	400	3.6	8.6	4.3	<0.001	0.98	100	100	0.0	n.s.	0.50
	Struct L=3	400	100	100	0.0	n.s.	0.50	7.6	7.6	0.0	n.s.	0.50
	Struct L=4	400	100	100	0.0	n.s.	0.50	2.5	2.5	0.0	n.s.	0.50
	Stoch	400	100	100	0.0	n.s.	0.50	0.3	0.3	0.0	<0.001	0.50
Te	MO L=2	400	100	100	0.0	n.s.	0.50	11.4	9.2	-2.5	<0.001	0.36
	MO L=3	400	100	100	0.0	n.s.	0.50	8.5	6.8	-1.7	<0.001	0.40
	MO L=4	400	100	100	0.0	n.s.	0.50	8.5	6.8	-1.7	<0.001	0.40
	Struct L=2	400	100	100	0.0	n.s.	0.50	6.8	3.9	-0.9	<0.001	0.38
	Struct L=3	400	100	100	0.0	n.s.	0.50	5.7	3.6	-0.7	<0.001	0.36
	Struct L=4	400	100	100	0.0	n.s.	0.50	4.9	2.7	-0.6	<0.001	0.32
	Stoch	400	100	100	0.0	n.s.	0.50	6.1	4.7	-1.5	<0.001	0.39
HS	MO L=2	416	100	100	0.0	n.s.	0.50	5.6	5.6	0.0	n.s.	0.50
	MO L=3	416	100	100	0.0	n.s.	0.50	3.7	3.7	0.0	n.s.	0.50
	MO L=4	416	100	100	0.0	n.s.	0.50	3.2	3.2	0.0	n.s.	0.50
	Struct L=2	416	40.3	33.3	-3.4	<0.001	0.24	100	100	0.0	n.s.	0.50
	Struct L=3	416	41.7	35	-3.2	<0.001	0.24	100	100	0.0	n.s.	0.50
	Struct L=4	416	43.1	36.7	-3.0	<0.001	0.24	100	100	0.0	n.s.	0.50
	Stoch	416	100	100	0.0	n.s.	0.50	2.6	2.6	0.0	n.s.	0.50

Table 10
S10: Damping-factor sensitivity, full results. The upper sub-table reports the Friedman omnibus tests: median per-SPL value at each damping factor and the Benjamini–Hochberg adjusted p-value across the three settings. Cells with $p_{BH} > 0.05$ are reported as n.s. The number of paired products n is the count of products with all three damping settings recorded. The lower sub-table reports the post-hoc Vargha–Delaney A_{12} effect sizes for the pairwise comparisons within each significant Friedman cell, with the magnitude classification (negligible / small / medium / large) used in the manuscript and the corresponding adjusted p-value.

Block	Metric	SPL	n	d=0.8	d=0.85	d=0.9	p_{BH}
Test generation	Edge coverage (%)	SVM	12	100.0	100.0	100.0	n.s.
		eM	23	100.0	100.0	100.0	n.s.
		EI	42	100.0	100.0	100.0	n.s.
		BAv2	498	100.0	100.0	100.0	n.s.
		SAS	2664	100.0	100.0	100.0	n.s.
		Svia	400	75.2	84.5	92.2	1.5×10^{-173}
		Te	400	100.0	100.0	100.0	4.5×10^{-48}
		HS	416	61.3	96.0	100.0	2.6×10^{-168}
	# test cases	SVM	12	2.0	2.0	2.0	n.s.
		eM	23	18.0	13.0	13.0	2×10^{-2}
		EI	42	38.0	33.0	38.0	n.s.
		BAv2	498	10.0	9.0	9.0	n.s.
		SAS	2664	37.0	33.0	36.0	4.1×10^{-42}
		Svia	400	4501.0	7346.0	12092.0	7.7×10^{-174}
		Te	400	13.0	13.0	12.5	1.5×10^{-2}
		HS	416	1.0	13.0	17.0	3.6×10^{-180}
	# test events	SVM	12	7.5	7.5	7.5	n.s.
		eM	23	86.0	80.0	87.0	3.1×10^{-2}
		EI	42	131.0	112.0	127.5	n.s.
		BAv2	498	35.0	35.5	34.5	n.s.
		SAS	2664	258.0	232.0	254.0	3.2×10^{-39}
		Svia	400	53452.0	95712.0	173964.0	7.7×10^{-174}
		Te	400	348.0	348.0	337.5	3.1×10^{-2}
		HS	416	45.0	618.0	869.0	1.7×10^{-180}
	Walk steps	SVM	12	14.5	14.5	14.5	n.s.
		eM	23	1430.0	369.0	271.0	1.1×10^{-7}
		EI	42	557.0	357.5	349.0	3.2×10^{-6}
		BAv2	498	439.0	284.0	206.5	1.7×10^{-22}
		SAS	2664	8111.0	4202.0	2956.0	$<10^{-300}$
		Svia	400	2000000.0	2000000.0	2000000.0	n.s.
		Te	400	93082.0	16049.0	4569.0	3.1×10^{-161}
		HS	416	1191640.0	1134905.0	294969.0	7.4×10^{-168}
Edge omission fault detection	Mutation score (%)	SVM	12	100.0	100.0	100.0	n.s.
		eM	23	100.0	100.0	100.0	n.s.
		EI	42	100.0	100.0	100.0	n.s.
		BAv2	498	100.0	100.0	100.0	n.s.
		SAS	2664	100.0	100.0	100.0	n.s.
		Svia	400	100.0	100.0	100.0	n.s.
		Te	400	100.0	100.0	100.0	1.4×10^{-47}
		HS	279	100.0	100.0	100.0	n.s.
	Penalized DC (%)	SVM	12	64.3	64.3	64.3	n.s.
		eM	23	21.9	27.4	27.4	n.s.
		EI	42	19.1	22.7	21.9	n.s.
		BAv2	498	46.7	47.2	47.1	n.s.
		SAS	2664	20.5	22.3	21.8	6.7×10^{-46}
		Svia	400	0.7	0.4	0.1	2.4×10^{-139}
		Te	400	5.6	5.8	5.9	n.s.
		HS	279	44.4	2.4	1.4	2×10^{-120}
Event omission fault detection	Mutation score (%)	SVM	12	100.0	100.0	100.0	n.s.
		eM	23	100.0	100.0	100.0	n.s.
		EI	42	100.0	100.0	100.0	n.s.
		BAv2	498	100.0	100.0	100.0	n.s.
		SAS	2664	100.0	100.0	100.0	n.s.
		Svia	400	100.0	100.0	100.0	n.s.
		Te	400	100.0	100.0	100.0	n.s.
		HS	279	100.0	100.0	100.0	n.s.
	Penalized DC (%)	SVM	12	57.1	57.1	57.1	n.s.
		eM	23	11.5	17.0	17.9	n.s.
		EI	42	6.3	6.3	5.5	n.s.
		BAv2	498	43.8	44.1	44.1	1×10^{-2}

Table 11
S11: Multi-seed stability of the stochastic baseline across 10 seeds (RQ3[b]). The coefficient-of-variation (CV) is computed per product across seeds; the table reports the per-SPL distribution summary. Bands are very_stable (median CV $< 5\%$), stable (5–10%), and notable_variance ($> 10\%$).

Operator	SPL	Metric	n	Median CV (%)	P25 (%)	P75 (%)	IQR (%)	Max (%)	Band
Edge omission	SVM	Unpenalized DC (%)	12	0.0	0.0	0.0	0.0	11.2	very_stable
		Mutation score (%)	12	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	12	0.0	0.0	0.0	0.0	11.2	very_stable
	eM	Unpenalized DC (%)	23	33.9	28.3	41.2	12.9	55.1	notable_variance
		Mutation score (%)	23	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	23	33.9	28.3	41.2	12.9	55.1	notable_variance
	EI	Unpenalized DC (%)	42	25.8	18.3	33.3	15.0	48.2	notable_variance
		Mutation score (%)	42	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	42	25.8	18.3	33.3	15.0	48.2	notable_variance
	BAv2	Unpenalized DC (%)	498	4.5	0.0	7.9	7.9	17.1	very_stable
		Mutation score (%)	498	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	498	4.5	0.0	7.9	7.9	17.1	very_stable
	SAS	Unpenalized DC (%)	2664	17.5	14.5	21.3	6.7	40.1	notable_variance
		Mutation score (%)	2664	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	2664	17.5	14.5	21.3	6.7	40.1	notable_variance
	Svia	Unpenalized DC (%)	400	101.7	93.1	108.1	15.0	166.6	notable_variance
		Mutation score (%)	400	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	400	101.7	93.1	108.1	15.0	166.6	notable_variance
	Te	Unpenalized DC (%)	400	18.3	13.4	21.8	8.4	49.2	notable_variance
		Mutation score (%)	400	0.0	0.0	0.0	0.0	0.6	very_stable
		Penalized DC (%)	400	18.3	13.4	21.8	8.4	49.2	notable_variance
	HS	Unpenalized DC (%)	416	11.8	9.5	15.0	5.5	23.8	notable_variance
		Mutation score (%)	416	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	416	11.8	9.5	15.0	5.5	23.8	notable_variance
Event omission	SVM	Unpenalized DC (%)	12	0.0	0.0	0.0	0.0	6.8	very_stable
		Mutation score (%)	12	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	12	0.0	0.0	0.0	0.0	6.8	very_stable
	eM	Unpenalized DC (%)	23	43.9	38.9	48.7	9.9	76.4	notable_variance
		Mutation score (%)	23	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	23	43.9	38.9	48.7	9.9	76.4	notable_variance
	EI	Unpenalized DC (%)	42	28.6	21.7	36.1	14.3	57.8	notable_variance
		Mutation score (%)	42	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	42	28.6	21.7	36.1	14.3	57.8	notable_variance
	BAv2	Unpenalized DC (%)	498	4.6	0.0	8.3	8.3	22.1	very_stable
		Mutation score (%)	498	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	498	4.6	0.0	8.3	8.3	22.1	very_stable
	SAS	Unpenalized DC (%)	2664	22.5	18.6	26.5	7.9	52.4	notable_variance
		Mutation score (%)	2664	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	2664	22.5	18.6	26.5	7.9	52.4	notable_variance
	Svia	Unpenalized DC (%)	400	101.7	92.8	108.1	15.3	158.6	notable_variance
		Mutation score (%)	400	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	400	101.7	92.8	108.1	15.3	158.6	notable_variance
	Te	Unpenalized DC (%)	400	18.0	13.4	21.7	8.4	49.2	notable_variance
		Mutation score (%)	400	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	400	18.0	13.4	21.7	8.4	49.2	notable_variance
	HS	Unpenalized DC (%)	416	11.8	9.5	15.0	5.5	23.8	notable_variance
		Mutation score (%)	416	0.0	0.0	0.0	0.0	0.0	very_stable
		Penalized DC (%)	416	11.8	9.5	15.0	5.5	23.8	notable_variance

Table 12
S12: RQ3[a] fault-detection results. Mutation score and penalized detection cost on edge and event omission for the original RQ1 Random Walk test sequences (single seed $42+productID$, damping 0.85). Stochastic baseline rows here are single-seed medians; the corresponding manuscript tables (Tables **??** and **??**) use the multi-seed median (RQ3[b]).

Approach	L	**SVM**	**eM**	**El**	**BAv2**	**SAS**	**Svia**	**Te**	**HS**
	2	100	100	100	100	100	100	100	100
Model Once, Generate Any	3	100	100	100	100	100	100	100	100
	4	100	95.2	100	91.5	100	100	100	100
	2	100	35.3	100	100	66.7	3.57	100	40.3
Structural baseline	3	100	76.2	100	100	85.3	100	100	41.7
	4	100	82.4	100	100	86.4	100	100	43.1
Stochastic baseline	—	100	100	100	100	100	100	100	100